

Tushar Tikia

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PROFESSIONAL SUMMARY

Software Development Engineer aspirant with strong expertise in full-stack development and AI-integrated systems. Experienced in designing scalable backend architectures, RESTful APIs, authentication systems, adaptive algorithms, and multi-agent AI workflows. Strong foundation in Data Structures, problem-solving, and system-oriented thinking.

EDUCATION

University School of Information, Communication & Technology, GGSIPU <i>Master of Computer Application, Software Engineering</i>	2025 – 2027 <i>Delhi, India</i>
Institute of Innovation in Technology and Management, GGSIPU <i>Bachelor of Computer Application</i>	2022 – 2025 <i>Delhi, India</i>

TECHNICAL PROJECTS

AI Interview Preparation Platform (Multi-Agent AI Architecture) | GitHub Repository

- Architected and developed a production-style full-stack AI platform with 15 modular components and 25+ REST API endpoints.
- Designed a centralized AI Orchestrator coordinating 6 specialized AI agents for question generation, evaluation, emotion analysis, speech analysis, adaptive difficulty, and feedback generation.
- Implemented role-based authentication (JWT + bcrypt) supporting user, admin, recruiter, and mentor access control.
- Engineered adaptive difficulty logic using performance trends, emotional state signals, and response-time heuristics.
- Developed real-time webcam monitoring and emotion tracking across 7 states with structured session logging.
- Integrated speech-to-text processing with filler word detection and answer clarity analysis.
- Built coding assessment module with structured test case validation and answer evaluation.
- Designed scalable MongoDB schema architecture (7 collections) with proper data modeling and relational referencing.
- Created analytics dashboard using Redux Toolkit + Chart.js for performance trends, radar skill visualization, and peer comparison.

Grocers Guru – Online Grocery E-Commerce (MERN Stack) | GitHub Repository

- Developed scalable e-commerce application with modular backend services and secure authentication workflows.
- Implemented cart management, order lifecycle handling, and payment integration logic.
- Designed admin inventory management dashboard with protected routes and role-based authorization.
- Structured RESTful API architecture for maintainable and extensible backend design.

TECHNICAL SKILLS

Programming: JavaScript (ES6+), C++
Frontend: React.js, Redux Toolkit, Tailwind CSS, Chart.js, Responsive UI Design
Backend: Node.js, Express.js, RESTful APIs, Middleware Architecture
Database: MongoDB, Mongoose, NoSQL Schema Design
AI Systems: OpenAI API Integration, Multi-Agent Architecture, Adaptive Algorithms
Core Concepts: Data Structures & Algorithms, OOP, Authentication, Role-Based Access Control
Tools: Git, GitHub, Postman, Axios, VS Code

CERTIFICATIONS

- **AWS Solution Architecture Job Simulation – Amazon Web Services:** Hands-on cloud system design simulation covering scalable architecture and reliability.
- **Full Stack Web Development – DevTown:** Built MERN stack applications focusing on backend API development and frontend responsiveness.
- **Gen AI Study Jams & Prompt Design – Google Cloud:** Explored prompt engineering and generative AI workflows.
- **Data Structures in C++ – Coding Ninjas:** Covered recursion, trees, graphs, and complexity optimization techniques.